

IDA ZHAO

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SUMMARY

Software engineer with experience building scalable backend systems, AI-powered automation tools, and data-driven platforms across fintech, e-commerce, and enterprise environments. Proficient in Python, TypeScript, C/C++, JavaScript, SQL, Snowflake, AWS, and GCP, with hands-on experience delivering production-grade monitoring systems, automated data pipelines, and distributed web applications in agile teams. Proven ability to quickly learn new technologies, independently own end-to-end engineering projects, and improve system reliability and efficiency through automation and data-driven solutions while collaborating across cross-functional teams.

EXPERIENCE

Software Engineering Intern - Data Engineering

BitGo | January 2026 - April 2026

- Improved blockchain balance reconciliation reliability by redesigning Node Balance Snapshot with a unified blockchain API integration framework and AI-assisted chain onboarding tooling using Claude, reducing USD balance delta from \$7.7B to \$21K while achieving a 99.96% reconciliation pass rate.
- Developed an AI-driven release automation system that summarized pull request changes using a custom memory bank workflow and distributed release updates through Slack integrations.
- Demonstrated strong adaptability and collaboration by contributing across multiple teams and quickly learning new technologies to deliver impactful solutions on critical engineering initiatives.

Data Analyst

Ford | January 2025 - August 2025

- Improved a high-impact vehicle-level feature usage report by redesigning product logic for modularity and automation that allowed seamless integration for new features and reduced feature onboarding time by 300%.
- Used Elasticsearch API to build a centralized, user-friendly data source, and cutting manual effort by 90%.
- Developed and maintained data workflows using Python, SQL, DBT and BigQuery, SSMS, Power BI and Looker Studio to support data-driven decision-making across business teams.

Web Developer & Data Analysis Assistant

Wave Mena | May 2024 - August 2024

- Built and delivered 7 responsive websites using HTML, CSS, JavaScript, and JSON, with a focus on data-driven UI/UX.
- Automated data maintenance workflows for projects, reducing manual reporting effort to 10%.

Front-End Developer

Mi BOWL | September 2023 - December 2023

- Enhanced Shopify-based e-commerce platform with HTML, CSS, and third-party apps.
- Automated order management by developing a Python script to process CSV files, improved operational efficiency by 60%.

PROJECTS

ChamberCrawler3000 (CC3k)

Tools: C++

- Engineered a text-based game in C++, emphasizing efficient memory usage and user interaction.
- Utilized robust algorithms to handle large data sets and optimize performance.

VI Markets Online Broker Website

Tools: HTML, CSS, JSON, JavaScript

- Led the development of a comprehensive brokerage platform, incorporating client-specific features and automation.
- Delivered the project within 4 weeks, enhancing site performance and user experience.

Node Balance Snapshot (NBS)

Tools: TypeScript, REST APIs, Kafka, Grafana, Snowflake, Claude

- Redesigned blockchain balance reconciliation architecture with a unified multi-chain API integration framework, improving monitoring, validation, and reconciliation reliability across chains.
- Built AI-assisted chain integration tooling and proactive alerting systems, increasing reconciliation accuracy to 99.96%.

SKILLS

- Programming Languages:** Python, C, C++, TypeScript, SQL, R, HTML, CSS, JavaScript, Bash, Kotlin
- Data & Cloud Technologies:** Pandas, NumPy, Matplotlib, DBT, BigQuery, Snowflake, Kafka, Grafana, AWS, GCP, Power BI
- Developer Tools & Platforms:** Git, Claude, Docker, VS Code, Postman, Kubernetes, Jira, Confluence, Figma, Linux, SSMS

EDUCATION

Bachelor of Computer Science, Co-op | University of Waterloo

September 2022 - Present

- GPA:** 3.95/4.0
- Relevant Coursework:** Object-Oriented Software Development, Software Design, Computer Organization, Algorithms and Complexity, Data Structures, Database Management, Machine Learning, Optimization, Combinatorics
- Awards & Scholarships:** University of Waterloo President's Scholarship of Distinction, 2023 Thiruchelvam Scholarship for Women in Technology, 2022 Euclid Contest Student Honour Roll (top 2%), 2022 Fermat Contest Regional Champion (Nova Scotia Provincial Champion)